

(NS-R) - formalism: $\{x^M, \chi_A^M\}, A=1,2$

- χ_A^M is NOT a spinor of the $M=1, \dots, D-1$ Background field.

on M^D : χ_A^M is a VECTOR

* χ_A^M is a spinor on Σ_g and transforms under $\text{Spin}(2)$ on its A -index.

This is $(1,1)$ -Susy. Can be constructed by reducing $(2,2)$ -theories

→ $(1,1)$: θ^+, θ^- are real up to phases

Supermultiplet (chiral): $\{x^M, \chi_A^M, N^M\}$

$$\int d^2\zeta \left(-\frac{1}{2} \partial_\alpha x^M \partial^\alpha x^N - \frac{i}{2} \bar{\chi}^M \not{\partial} \chi^N + \frac{1}{2} N^M N^N \right) \eta_{MN}$$

→ Coupling this theory to gravity amounts to:

$$S = \frac{1}{2\pi\alpha'} \int d^2\zeta e \left\{ -\frac{1}{2} e_a^\alpha \partial_\alpha x^M e^{a\beta} \partial_\beta x^N - \frac{i}{2} \bar{\chi}^M \not{\partial} \chi^N + \frac{1}{2} N^M N^N \right. \\ \left. + \frac{i\kappa}{2} \bar{\psi}_\alpha \not{\partial} x^M \gamma^\alpha \chi^N - \frac{\kappa^2}{16} \bar{\psi}_\alpha \gamma^\beta \gamma^\alpha \bar{\chi}^M \chi^N \right\} \eta_{MN} \\ \equiv \int d^2\zeta \mathbb{L}$$

which has the symmetries

$$\begin{aligned} \delta e_a^\alpha &= f^B \partial_B e_a^\alpha + \partial_a f^B e_B^\alpha \\ \delta \psi_{aA} &= f^B \partial_B \psi_{aA} + \partial_a f^B \psi_{BA} \\ \delta x^M &= f^B \partial_B x^M \\ \delta \chi_A^M &= f^B \partial_B \chi_A^M(x) \\ \delta N^M &= f^B \partial_B N^M \end{aligned}$$

Diffeomorphism invariance

$$\mu: \longrightarrow \eta_{MN} \text{ on } M^D$$

$$a: \longrightarrow \eta_{ab} \text{ on } \Sigma_g \cong M^2$$

$$\alpha: \longrightarrow g_{\alpha\beta} \text{ on } \Sigma_g \text{ via}$$

$$g_{\alpha\beta} = e_a^\alpha \eta_{ab} e_b^\beta$$

$$e \equiv \det(e_a^\alpha) \quad \not{\partial} \equiv \gamma^\alpha e_a^\alpha \partial_\alpha \\ \zeta^\alpha \longrightarrow \zeta^\alpha + f^\alpha$$

$$\begin{aligned} X^M(\xi') - X^M(\xi) &= X^M(\xi) + f^\alpha \partial_\alpha X^M - X^M(\xi) \\ &= f^\alpha \partial_\alpha X^M = \delta X^M \end{aligned}$$

(i) 2D Lorentz invariance

$$\begin{cases} \delta e_a^\alpha = -\omega^a_b e_a^\alpha \\ \delta X_A^M = \frac{1}{4} (\omega^{ab} \gamma_{ab})_A^B X_B^M \\ \delta \psi_{\alpha A} = \frac{1}{4} (\omega^{ab} \gamma_{ab})_A^B \psi_{\alpha B} \end{cases}$$

(ii) SUSY

$$\delta X^M = i \bar{\epsilon} \gamma^M \chi, \quad \delta \chi^M = (\hat{D} X^M + N^M) \epsilon$$

$$\delta N^M = i \bar{\epsilon} \hat{D} \chi^M$$

$$\delta e_a^\alpha = i \kappa \bar{\epsilon} \gamma^\alpha \psi_a, \quad \delta \psi_a = \frac{2}{\kappa} D_a \epsilon$$

where

$$\hat{D}_a X^M = \partial_a X^M - \frac{i\kappa}{2} \bar{\psi}_a \gamma^M \chi$$

$$\hat{D}_\beta \chi^M = (D_\beta \chi^M - \frac{\kappa}{2} \hat{D} \chi^M + N^M) \psi_\beta$$

$$\hat{D}_\beta \chi^M = (\partial_\beta + \frac{1}{4} \omega_{\beta ab} \gamma^{ab}) \chi^M$$

$$D_\beta \epsilon = (\partial_\beta + \frac{1}{4} \omega_{\beta ab} \gamma^{ab}) \epsilon$$

Weyl Symmetry:

$$\delta e_a^\alpha = \Lambda e_a^\alpha, \quad \delta \psi_a = \frac{1}{2} \Lambda \psi_a$$

$$\delta X^M = 0, \quad \delta \chi^M = -\frac{1}{2} \Lambda \chi^M$$

$$\delta N^M = -\Lambda N^M$$

Local S-symmetry: $\delta \psi_a = \gamma_a \zeta$

where ζ_A is a Majorana spinor which depends on ξ^α in an arbitrary way. All other fields are inert.

Poincare: $\delta X^M = -\Lambda_N^M X^N + a^M, \quad \delta \chi^M = -\Lambda_N^M \chi^N$ etc...

* Poincare is a Global symm.

E.O.M:

$$\chi^M: \partial_\alpha (e e_a^\alpha e^{ab} \partial_\beta X^M) - \frac{i\kappa}{2} \partial_\beta (e e^{\beta a} \bar{\psi}_a \gamma^M \gamma^a \chi) = 0$$

$$\chi^M: \partial \chi^M - \frac{\kappa}{2} \gamma^a \partial X^M \psi_a - \frac{i\kappa^2}{8} \chi^M \bar{\psi}_a \gamma^B \gamma^a \psi_B$$

Since χ^M are Majorana, then

$$\bar{\chi} \gamma^{M_1} \dots \gamma^{M_P} \chi = (-1)^P \bar{\chi} \gamma^{M_P} \dots \gamma^{M_1} \chi$$

then

$$\frac{i\kappa}{2} \bar{\psi}_\alpha \gamma^\beta \partial_\beta x^\mu \gamma^\alpha \chi^\mu = \frac{i\kappa}{2} \chi^\mu \gamma^\alpha \gamma^\beta \psi_\alpha \partial_\beta x^\mu$$

$$= \frac{i\kappa}{2} \bar{\chi}^\mu \gamma^\alpha \partial_\alpha x^\mu \psi_\alpha$$

we then use $\delta \mathcal{L} / \delta x^\mu = 0$ to find the above result.

→ Pretty sure this is -1 and not $(-1/2)$

* $N^\mu = 0$ (E.O.M)

$$\psi_{\alpha A}: \frac{\kappa}{2} \partial_\alpha x^\mu \gamma^\alpha \chi^\mu + \frac{i\kappa^2}{8} \gamma^\beta \gamma^\alpha \psi_\beta \bar{\chi}^\mu \chi_\mu$$

→ Book has $\frac{1}{8}$ instead of $\frac{1}{8} \dots$ Not sure about this

E.O.M of e_α^a is fucked... see text

* Since the Superstring is extended we also must have B.C.'s:

From $\delta \bar{X}^\mu$ we know:

$$\bar{\pi}^\sigma \equiv \frac{\delta \mathcal{A}}{\delta (\partial_\sigma x^\mu)} \delta x^\mu \text{ must vanish}$$

$$\text{as well as } \frac{\delta \mathcal{A}}{\delta (\partial_\sigma x_A^\mu)} \delta x_A^\mu$$

Both must vanish on the ends of the string
i.e. at 0 and π

Open string: 0 and π

Closed string: $-\pi$ and π

* Now make gauge choices:

e_α^a is subject to γ -local transformations

($\Rightarrow$ Lorentz, two diffs, Weyl

$$g_{\alpha\beta} = e_\alpha^a \eta_{ab} e_\beta^b$$

We Gauged $g_{\alpha\beta}$ to the form

$$\gamma_{ab} = M_{ab} e^{-\phi}$$

and so $e^a_\alpha M_{ab} e^b_\beta = M_{\alpha\beta} e^{2\phi}$

$\therefore$

$$\tilde{e}^a_\alpha M_{ab} \tilde{e}^b_\beta = M_{\alpha\beta}, \quad \tilde{e} = e^{-\phi} e$$

so $\tilde{e}$ belongs to $SO(1,1)$

$$\tilde{e}^a_\alpha = \delta^a_\alpha = e^{-\phi} e^a_\alpha$$

$$\therefore e^a_\alpha = \delta^a_\alpha e^\phi$$

Using Weyl we can then conclude

$$e^a_\alpha = \delta^a_\alpha$$

ψ_α : has 4 components

Subject to 4 local fermi.

$$\begin{pmatrix} 2 \text{ susy} \\ 2 \text{ S-susy} \end{pmatrix}$$

To do this we decompose ψ_α as follows:

$$\psi_\alpha = (\delta^B_\alpha - \frac{1}{2} \gamma_\alpha \gamma^B) \psi_B + \frac{1}{2} \gamma_\alpha \gamma^B \psi_B$$

$$\gamma_\alpha \gamma^B = -\gamma^B \gamma_\alpha + 2 \delta^B_\alpha \mathbb{1}$$

$$\therefore -\frac{1}{2} (\gamma_\alpha \gamma^B) = -\frac{1}{2} \gamma_\alpha \gamma^B = \frac{1}{2} \gamma^B \gamma_\alpha - \delta^B_\alpha$$

$$\begin{aligned} \psi_\alpha &= (\delta^B_\alpha + \frac{1}{2} \gamma^B \gamma_\alpha - \delta^B_\alpha) \psi_B + \frac{1}{2} \gamma_\alpha \gamma^B \psi_B \\ &= \frac{1}{2} \gamma^B \gamma_\alpha \psi_B + \frac{1}{2} \gamma_\alpha (\gamma^B \psi_B) \\ &\equiv \psi_\alpha + \gamma_\alpha \phi \end{aligned}$$

$$\psi_\alpha \equiv \frac{1}{2} \gamma^B \gamma_\alpha \psi_B \quad (\text{traceless})$$

$$\phi \equiv \frac{1}{2} \gamma^B \psi_B$$

$$(\gamma^\alpha)_A{}^B (\psi_a)_B = 0 \quad \left(\text{there can only be } \mathcal{O}(\gamma^2) \right. \\ \left. \text{in the worldsheet} \right)$$

ψ_a then has $(2 \times 2) - 2 = 2$ components

ψ_a is orthogonal w.r.t. $(\phi, \phi) \equiv \int d^2\sigma \bar{\phi}^a \psi_a$

The same decomp can be done for its

susy variation

$$\delta_\epsilon \psi_\alpha = 2 D_\alpha \epsilon$$

$$\equiv 2(\pi\epsilon)_\alpha + \gamma_\alpha \gamma^\beta D_\beta \epsilon$$

where the operator $(\pi\epsilon)_\alpha$ is defined as

$$(\pi\epsilon)_\alpha \equiv (\delta_\alpha^\beta - \frac{1}{2} \gamma_\alpha \gamma^\beta) D_\beta \epsilon = \frac{1}{2} \gamma^\beta \gamma_\alpha D_\beta \epsilon$$

$\pi: (\text{Spin } 1/2) \longrightarrow (\gamma\text{-traceless spin } 3/2 \text{ field})$

$\therefore$ locally we may write

$$\psi_\alpha = \gamma^\beta \gamma_\alpha D_\beta \mathcal{M}$$

for some spinor $\mathcal{M}$

$\Rightarrow$ in light-cone coordinates: $\psi_+ = \begin{pmatrix} 2D_+ \mathcal{M} \\ 0 \end{pmatrix}, \psi_- = \begin{pmatrix} 0 \\ 2D_- \mathcal{M} \end{pmatrix}$

$$\psi_\alpha = \gamma_\alpha \psi + \gamma^\beta \gamma_\alpha D_\beta \mathcal{M}$$

$$= \gamma_\alpha \psi - \gamma_\alpha \gamma^\beta D_\beta \mathcal{M} + 2\delta_\alpha^\beta D_\beta \mathcal{M}$$

$$\therefore \boxed{\psi_\alpha = 2D_\alpha \mathcal{M} + \gamma_\alpha (\psi - \not{D}\mathcal{M})}$$

From susy we note

$$\psi_\alpha - \gamma_\alpha (\psi - \not{D}\mathcal{M}) = \delta_\epsilon \psi_\alpha$$

$\therefore$ this term vanishes when we apply a susy to ψ_α .

$$\psi_\alpha = \gamma_\alpha (\psi - \not{D}\mathcal{M})$$

Can further use S-susy by defining

$$\xi = (\psi - \not{D}\mathcal{M})$$

so $\psi_\alpha = \delta_s \xi$ which will vanish as well.

i.e. ψ_α is built by exact forms of the symm. algebra of the theory.

$$\psi_\alpha = 0$$

$\rightarrow$ Further, note that with the form

$\psi_\alpha = \gamma_\alpha (\psi - \not{D}\mathcal{M})$ Plugging this back into $\mathcal{L}$ yields a zero ψ_α containing terms as they are $\mathcal{O}(\theta^3) = 0$.

Choices:

$\psi_\alpha = \gamma_\alpha (\psi - \not{D}\mathcal{M})$

$$\left\{ \begin{aligned} \psi_\alpha &= \psi_\alpha (4 - \not{D}) \chi \\ e_\alpha^a &= \delta_\alpha^a e^a \end{aligned} \right\}$$

$\leadsto$ E.O.M become

$$\begin{aligned} \chi^\mu: \partial_\alpha (e e_\alpha^a e^{c\beta} \partial_\beta \chi^\mu) &= 0 \\ &= \partial_\alpha (e^{(4/3)} \partial_\beta \chi^\mu \delta_\alpha^a \delta^{c\beta} \partial_\beta \chi^\mu) = 0 \\ \therefore \partial_\alpha \partial^\alpha \chi^\mu &= 0. \end{aligned}$$

$$\chi^\mu: \not{D} \chi^\mu = 0$$

The e_α^a E.O.M is simply the E.M. - tensor

$$\text{Since } \frac{1}{e} \frac{\delta \mathcal{L}}{\delta g^{\alpha\beta}} = 0$$

and ψ_μ give constraints ... interesting.

e_α^a :

$$\partial_\alpha \chi^\mu e_\alpha^a \partial_\beta \chi^\mu + \frac{i}{2} \bar{\chi}^\mu \gamma^\alpha \partial_\alpha \chi^\mu + e_\alpha^a \text{tr} \left(-\frac{1}{2} e e^a \partial_\alpha \chi^\mu \partial_\beta \chi^\mu \right)$$

$$\begin{aligned} T_{\alpha\beta} &= \partial_\alpha \chi^\mu \partial_\beta \chi^\mu \eta_{\mu\nu} + \frac{i}{2} \bar{\chi} \gamma_\beta \partial_\alpha \chi^\mu \eta_{\mu\nu} \\ &\quad - \frac{\eta_{\alpha\beta}}{2} (\partial_\gamma \chi^\mu \partial^\gamma \chi^\mu + i \bar{\chi}^\mu \not{D} \chi^\mu) \eta_{\mu\nu} = 0 \end{aligned}$$

and

$$J^\alpha \equiv \not{D} \chi^\mu \partial^\alpha \chi^\mu \eta_{\mu\nu} = 0$$

* This super current is a susy dual to the E.M. tensor

$$T_{\alpha\beta} = \frac{2\pi}{e} \frac{\delta \mathcal{A}}{\delta e_\alpha^a} e_{\alpha a} \quad | \quad J^\alpha \text{ is then}$$

$$\begin{aligned} \delta \mathcal{A} &= \frac{1}{2\pi} \int d^2\sigma e \left(i \frac{\delta \mathcal{A}}{\delta \bar{\chi}^a} \delta \bar{\chi}^a \right) = 0 \\ &= \frac{1}{2\pi} \int d^2 e i \delta \bar{\chi}^a J_a \end{aligned}$$

$$\therefore \frac{e}{2\pi} = \frac{\delta \mathcal{A}}{i \delta \bar{\chi}^a} \Rightarrow J_a = \frac{2\pi}{e} \frac{\delta \mathcal{A}}{\delta \bar{\chi}^a}$$

This Normalization is such that:

$$\delta_e J_\alpha = \frac{1}{2} T_{\alpha\beta} \gamma^\beta \epsilon$$

* $T_{\alpha\beta}$ is Not symmetric, must include the term

(which is anti-sym)

$$\frac{i}{4} \{ (\bar{\chi}^M \gamma_B \partial_\alpha \chi^N) + (\bar{\chi}^M \gamma_\alpha \partial_B \chi^N) \} \eta_{MN}$$

which vanishes on-shell as $\epsilon^{\alpha\beta} \gamma_\beta = \gamma_5 \gamma^\alpha$

$$= -\frac{i}{4} \bar{\chi}^M \epsilon_{\alpha\beta} \partial_5 \partial^\alpha \chi^N \eta_{MN}$$

$$T_{\alpha\beta} = \partial_\alpha \chi^\mu \partial_\beta \chi_\mu + \frac{i}{4} (\bar{\chi}^M \gamma_\beta \partial_\alpha \chi^N + \bar{\chi}^M \gamma_\alpha \partial_\beta \chi^N) \eta_{MN} \\ - \eta_{MN} \frac{1}{2} (\partial_\alpha \chi^\mu \partial_\beta \chi^\nu + \frac{i}{2} \bar{\chi}^M \partial^\alpha \chi^N) \eta_{\alpha\beta}$$

→ Conserved, symm and traceless

$$(\gamma_5)_A^B J_B^S = 0$$

The Open-Superstring:

Gauge choices: $\left(\begin{array}{l} e_\alpha^\mu = \delta_\alpha^\mu e^\mu \\ \psi_\alpha = \gamma_\alpha \psi + \varphi_\alpha \end{array} \right)$

$$\frac{\delta A}{\delta \partial \chi_A^\mu} \delta \chi_A^\mu \Big|_0^\pi = 0$$

$$\left(\frac{i}{2} \bar{\chi}_\mu \gamma^0 \delta \chi^\mu \right) \Big|_0^\pi = 0$$

Recall for the Bosonic-string we required this to vanish on $\int d\sigma d\tau \rightarrow \int_0^\pi d\sigma$
or, $\gamma^0 = \gamma^1$. This is the term of interest

Boundary Condition: $\bar{\chi}^M \gamma^1 \delta \chi_M \Big|_0^\pi = 0$

Notation: $\chi_A^\mu \equiv \begin{pmatrix} \chi_+^\mu \\ \chi_-^\mu \end{pmatrix}$; $\gamma^B \equiv (\gamma^0, \gamma^1)$

$$\gamma_2 \equiv \gamma^0 \gamma^1, \quad C \equiv -\gamma^0$$

$$\bar{\chi}^0 \equiv -\chi^\dagger \gamma^0 \quad (\text{Dirac}) \quad \left| \begin{array}{l} \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{array} \right.$$

$$\bar{\chi}^M \equiv \chi^T C \quad (\text{Majorana}) \quad \left| \begin{array}{l} \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \end{array} \right.$$

$$\gamma^B = \left(\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right)$$

$$\epsilon_{\alpha\beta} \equiv \epsilon^{\alpha\beta} \quad \left| \begin{array}{l} \epsilon_{12} = 1 \\ \epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \end{array} \right.$$

$$\gamma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma^3$$

$$\therefore \gamma_{\alpha\beta}^0 \equiv \epsilon_{\alpha\beta}$$

$$* (\chi_-^\mu \delta \chi_{\mu-} - \chi_+^\mu \delta \chi_{\mu+}) \Big|_0^\pi = 0$$

Ends of string are not causally connected i.e. we cannot have this eq. be satisfied w/ mixed coords

$$\chi_-^\mu \delta \chi_-^\mu = \chi_+^\mu \delta \chi_+^\mu \quad \text{separately at } \sigma=0 \quad \sigma=\pi$$

- χ_A^μ obeys a first order equation and if it were subject to 4 B.C.'s it would not possess interesting solⁿs (it is overdetermined.)

$$\chi_+^\mu(0) = \tau(0) \chi_-^\mu(0)$$

$$\chi_+^\mu(\pi) = \tau(\pi) \chi_-^\mu(\pi)$$

Provided that

$$\chi_-^\mu \delta \chi_-^\mu = \tau \chi_-^\mu \delta (\tau \chi_-^\mu)$$

$$= \tau^2 \chi_-^\mu \delta \chi_-^\mu$$

$$\tau^2(0) = \tau^2(\pi) = 1$$

- ? Since χ^μ is spinorial, we can choose $\chi_+^\mu(0) = \chi_-^\mu(0)$, $\tau(0) = 1$

→ At $\sigma = \pi$, we are left with two choices

$$\chi_+^\mu(\pi) = \chi_-^\mu(\pi)$$

(R) Ramond

$$\chi_+^\mu(\pi) = -\chi_-^\mu(\pi)$$

(NS) Neveu-Schwarz

R and NS Sectors

Upon Quantization:

(R): leads to space-time fermions

(NS): Spacetime Bosons

→ Use lightcone coordinates on Σ_g :

$$\xi^+ \equiv \tau + \sigma, \quad \xi^- \equiv \tau - \sigma$$

$$S^+ = S^0 + S^1, \quad S^- = S^0 - S^1$$

$$g_{\alpha\beta}(\xi) = \frac{\partial x^\mu}{\partial \xi^\alpha} \frac{\partial x^\nu}{\partial \xi^\beta} \eta_{\mu\nu}$$

$$= \frac{\partial x^0}{\partial \xi^\alpha} \frac{\partial x^0}{\partial \xi^\beta} (-1) + \frac{\partial x^1}{\partial \xi^\alpha} \frac{\partial x^1}{\partial \xi^\beta}$$

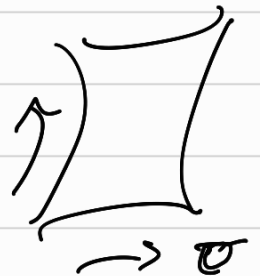
$$(\xi^+ + \xi^-) \frac{1}{2} = \tau, \quad \sigma = \frac{1}{2}(\xi^+ - \xi^-)$$

$$g_{\alpha\beta}(\xi) = \left(-\frac{1}{4}\right) \delta_{\alpha\pm} \delta_{\beta\pm} + \frac{1}{4} \left(\pm \delta_{\alpha\pm}\right) \left(\pm \delta_{\beta\pm}\right)$$

$$g_{++} = -\frac{1}{4} + \frac{1}{4} = 0$$

$$g_{--} = 0$$

$$g_{+-}(\xi^\pm) = -\frac{1}{4} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$



$$g_{+-} = -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}$$

$$g_{-+} = -1/2$$

$$g_{\alpha\beta}(3-1) = -1/2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$g^{\alpha\beta} = -2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\therefore S_+ = \frac{1}{2}(S_0 + S_1), \quad S_- = \frac{1}{2}(S_0 - S_1)$$

$$\partial_+ \equiv \partial/\partial \xi^+ = \frac{1}{2}(\partial/\partial \tau + \partial/\partial \sigma)$$

$$\partial_- \equiv \partial/\partial \xi^- = \frac{1}{2}(\partial/\partial \tau - \partial/\partial \sigma)$$

* The constraints become

$$T_{++} = \partial_+ X^\mu \partial_+ X^\mu \eta_{\mu\nu} + \frac{i}{4} (\bar{\chi}^+ \gamma_+ \partial_+ \chi^+ + \bar{\chi}^+ \gamma_+ \partial_+ \chi^+) \eta_{\mu\nu}$$

$$+ \mathcal{H}_{++}() \leftarrow \text{variables}$$

$$T_{++} = 0 = \partial_+ X^\mu \partial_+ X_\mu + \frac{i}{2} \bar{\chi}^+ \gamma_+ \partial_+ \chi_\mu$$

$$T_{--} = 0 = \partial_- X^\mu \partial_- X_\mu + (i/2) \bar{\chi}^+ \gamma_- \partial_- \chi_\mu$$

Note, T is traceless so,

$$\eta^{\alpha\beta} T_{\alpha\beta} = 0$$

$$(-2)(T_{-+} + T_{+-}) = 0$$

$$\text{But } T \text{ is symmetric so } T_{+-} = 0$$

$$\gamma_+ = \frac{1}{2}(\gamma_0 + \gamma_1) = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\gamma_- = \frac{1}{2}(\gamma_0 - \gamma_1) = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$\therefore$

$$\chi^\mu = (\chi_+, \chi_-)$$

$$= (\chi_-, \chi_+)$$

$$\bar{\chi}^\mu \gamma_\mu = (\chi_-, -\chi_+) \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \partial \chi_+ \\ \partial \chi_- \end{pmatrix} = (\chi_-, -\chi_+) \begin{pmatrix} 0 \\ \partial \chi_+ \end{pmatrix}$$

$$= -\chi_+ \partial_+ \chi_\mu$$

$$\bar{\chi}^\mu \gamma_- \partial_- \chi_\mu = -\chi_- \partial_- \chi_\mu$$

So,

$$T_{++} = \partial_+ X^\mu \partial_+ X_\mu - \frac{i}{2} \chi_+ \partial_+ \chi_\mu$$

$$T_{--} = \partial_- X^\mu \partial_- X_\mu - \frac{i}{2} \chi_- \partial_- \chi_\mu$$

$$S^\alpha = \partial X^\mu \gamma^\alpha \chi^\mu \eta_{\mu\nu}$$

$$S^+ = \partial X^\mu \gamma^+ \chi^\mu \eta_{\mu\nu}$$

$$S^- = \partial X^\mu \gamma^- \chi^\mu \eta_{\mu\nu}$$

$$J_A^+ = (\gamma^\alpha \partial_\alpha \chi^\mu \gamma^+ \chi^\nu)_A \quad \mathcal{M}_{\mu\nu} = (\gamma^\alpha)_A^B \gamma^+ \gamma^\alpha \chi_\mu^\nu \partial_\alpha \chi_\nu = 0$$

$$\gamma_+ = \mathcal{M}_+ - \gamma_- = -\frac{1}{2} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\therefore \gamma_- = \begin{pmatrix} 0 & 0 \\ -2 & 0 \end{pmatrix}$$

$$\gamma^+ = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$$

$$\therefore \begin{cases} \gamma^+ \gamma^+ = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = 0 \\ \gamma^- \gamma^- = 0 \end{cases}$$

$$\gamma^0 \gamma^+ = \begin{pmatrix} 0 & 0 \\ -2 & 0 \end{pmatrix} \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -4 \end{pmatrix}$$

$$\gamma^+ \gamma^- = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ -2 & 0 \end{pmatrix} = \begin{pmatrix} -4 & 0 \\ 0 & 0 \end{pmatrix}$$

$$J_A^+ = \gamma_- \partial_- \chi_\mu \gamma^+ \chi^\nu$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & -4 \end{pmatrix} \begin{pmatrix} \chi_+^\mu \\ \chi_-^\mu \end{pmatrix} \partial_- \chi_\nu$$

$$\Rightarrow J_A^+ = \begin{pmatrix} 0 \\ -4 \partial_- \chi_\mu \chi^\mu \end{pmatrix}$$

$$J_+^+ = 0$$

$$J_-^+ = -4 \partial_- \chi_\mu \chi^\mu$$

Similarly we find

$$J_-^- = 0$$

$$J_+^- = -4 \partial_+ \bar{\chi}_\mu \chi^\mu$$

* Extend range of σ from $[0, \pi] \rightarrow [-\pi, \pi]$

by

$$\bar{X}^\mu(\sigma) \equiv \begin{cases} \bar{X}^\mu(\sigma) & 0 \leq \sigma < \pi \\ \bar{X}^\mu(-\sigma) & -\pi \leq \sigma < 0 \end{cases}$$

$$\Phi^\mu(\sigma) \equiv \frac{1}{\alpha} \begin{cases} \chi_+^\mu(\sigma) & 0 \leq \sigma < \pi \\ \chi_-^\mu(-\sigma) & -\pi \leq \sigma < 0 \end{cases}$$

$$\rightarrow$$

NOT Gravitino

Already have B.C.: $\chi_+^\mu(0) = \chi_-^\mu(0)$

~> For Ramond B.C.'s: $\chi_+^\mu(\pi) = \chi_-^\mu(\pi)$

$$\Phi(\pi) = \frac{1}{\alpha} \cdot \chi_+^\mu(\pi) = \frac{1}{\alpha} \cdot \chi_-^\mu(\pi)$$

$$\Phi(-\pi) = \frac{1}{\alpha} \chi_-^\mu(-\pi)$$

$$\psi(-\pi) = \alpha \cdot \psi(\pi)$$

$$\therefore \boxed{\Psi(\pi) = \Psi(-\pi)} \quad (R)$$

For NS: $\chi_+^\mu(\pi) = -\chi_-^\mu(\pi)$

$$\Psi(\pi) = \frac{1}{\alpha'} \chi_+^\mu(\pi) = -\frac{1}{\alpha'} \chi_-^\mu(\pi)$$

$$\Psi(-\pi) = \frac{1}{\alpha'} \chi_+^\mu(\pi)$$

$$\therefore \boxed{\Psi(\pi) = -\Psi(-\pi)} \quad (NS)$$

Introduce ladder ops

$$\boxed{\alpha^\mu = \frac{1}{\sqrt{2\alpha'}} (\dot{x}^\mu + x'^\mu)}$$

Recall $2\partial_+ = \partial_\tau + \partial_\sigma$

$$\boxed{\alpha^\mu = \frac{2}{\sqrt{2\alpha'}} \partial_+ x^\mu = \frac{2}{\sqrt{2\alpha'}} \begin{cases} \partial_+ x^\mu & 0 \leq \sigma < \pi \\ \partial_- x^\mu & -\pi \leq \sigma < 0 \end{cases} = \frac{2}{\sqrt{2\alpha'}} \Delta x^\mu}$$

Since

$$\ddot{x}^\mu + \frac{d}{d\sigma}(x'^\mu(-\sigma)) = \ddot{x}^\mu - x'^{\mu'}(\sigma) = 2\partial_- x^\mu$$

* The E.M. - tensor constraints are equivalent to

$$\boxed{T^{\mu\nu} - \frac{1}{2}\eta^{\mu\nu}T = 0} \quad \text{for } -\pi < \sigma < \pi$$

• Super-current constraints:

$$\alpha^\mu \psi_\mu = 0$$

~> Each range $0 \leq \sigma < \pi$ and $-\pi \leq \sigma < 0$ yield two sets of eq's.

$$\boxed{x^\mu = x_0^\mu + \frac{1}{2} \sum_{n=-\infty, n \neq 0} e^{in\sigma} \alpha_n^\mu} \quad | \quad x_{-n}^\mu = x_n^\mu$$

$$\boxed{\alpha^\mu = \sum_{n=-\infty}^{\infty} e^{in\sigma} \alpha_n^\mu}$$

~> For the fermion field introduce modes as dictated by the B.C.'s (R) or (NS)

$$R \psi^\mu = \sum_{n \in \mathbb{Z}} d_n^\mu(\tau) e^{-in\sigma}$$

(R)

$$NS \psi^\mu = \sum_{n \in \mathbb{Z} + 1/2} b_n^\mu(\tau) e^{-in\sigma}$$

(NS)

R is obvious, for $r = n + 1/2$

$$\Psi_{(\pi)}^{\mu} = \sum_{n \in \mathbb{Z} + 1/2} b_r^{\mu}(\tau) e^{in\tau} e^{-i\pi/2}$$

$$= \sum b_r^{\mu}(1)(-1) = -\psi(-\pi)$$

$$\psi^{\mu}(-\pi) = \sum b_r^{\mu}(\tau) (1)(1) \Rightarrow$$

As expected:

$$\left. \begin{array}{l} \{r, s, t \dots \in \mathbb{Z}/2\} \\ \{m, n, p \in \mathbb{Z}\} \end{array} \right\} \not\chi = 0$$

From the Massless Dirac we have Weyl E.O.M:

$$\partial_- \chi_+ = 0 = \partial_+ \chi_-$$

Recall we took $\{\psi^{\mu}, \bar{\psi}^{\mu}\}$ to find

$$\alpha_n^{\mu}(\tau) = e^{in\tau} \alpha_n^{\mu}(0)$$

In much the same way we find

$$b_r^{\mu}(\tau) = e^{in\tau} b_r^{\mu}(0)$$

$$d_n^{\mu}(\tau) = e^{in\tau} d_n^{\mu}(0)$$

$\therefore$ Proof:

$$\partial_- \chi_+ = \frac{1}{2} (\partial_{\tau} - \partial_{\sigma}) \chi_+ = 0$$

$$\partial_{\tau} \chi_+ = \partial_{\sigma} \chi_+ \Rightarrow$$

$$\sum f_a^{\mu}(\tau) e^{-ia\sigma} = \sum f_a^{\mu}(-ia) e^{-ia\sigma}$$

$$\therefore f_a^{\mu}(\tau) = -ia f_a^{\mu}(\tau)$$

$$\ln(f) = -ia\tau + c$$

$$f_a^{\mu}(\tau) = c e^{-ia\tau}$$

$$f_a^{\mu}(0) = c \Leftrightarrow f_a^{\mu}(\tau) = f_a^{\mu}(0) e^{-ia\tau}$$

where $f = d$ or $f = b$ and $a = n$ or $a = r$

$$\Phi^{\mu} = \sum_{n \in \mathbb{Z}} d_n^{\mu}(0) e^{-in\tau}$$

(12)

$$\Psi^{\mu} = \sum_{r \in \mathbb{Z} + 1/2} b_r^{\mu}(0) e^{-ir\tau}$$

(NS)

$$\gamma^{\mu} = \sum_n \alpha_n^{\mu}(0) e^{-in\tau}$$

* Introduce Fourier Modes for the Constraints

$$L_n = \frac{1}{4\pi} \int_{-\pi}^{\pi} e^{in\sigma} d\sigma \{ \dot{\Phi}^{\mu} \dot{\Phi}_{\mu} - i \Phi^{\mu} \Delta \Phi^{\nu} \eta_{\mu\nu} \}$$

Recall these arise from

$$T = \sum L_n \dot{z}^n - u$$

and mapping $S^1 \times \mathbb{R} \rightarrow \mathbb{P}^2 \quad z, \bar{z} \rightarrow \tau, \sigma$

$$F_n = \frac{2}{2\pi} \int_{-\pi}^{\pi} e^{in\sigma} \partial^\mu \Psi_n d\sigma, \quad n=0, \pm 1, \dots \quad (12)$$

$$G_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{in\sigma} \partial^\mu \bar{\Psi}_n d\sigma, \quad n = \pm 1/2, \pm 3/2, \dots \quad (13)$$

* in the above Gauge $P^\mu = \dot{x}^\mu / 2\pi\alpha'$

$$\{x^\mu(\sigma), P^\nu(\sigma')\} = \delta(\sigma - \sigma') \eta^{\mu\nu}$$

For fermions $\chi_\pm^\mu$: this is more complicated

(i) Poisson for Grassmann odd objects

(ii) The canonical Momentum of $\chi_\pm^\mu$ is $(\frac{i}{4\pi\alpha'}) \chi_\pm^\mu$

$$\begin{aligned} P_\pm^\mu &= \frac{\partial \mathcal{L}}{\partial \partial_\tau \chi_\pm^\mu} = \left(-\frac{i}{4\pi\alpha'} \right) \bar{\Psi}^\mu \gamma^0 \\ &= \frac{i}{4\pi\alpha'} (-\chi^\mu) (\gamma^0)^2 \\ &= \frac{i}{4\pi\alpha'} (\chi_+^\mu, \chi_-^\mu) \end{aligned}$$

$\Rightarrow$ so χ^μ is its own momenta which we call $\tilde{\gamma}_\pm^\mu$

$$\tilde{\gamma}_\pm^\mu - \frac{i}{2\pi\alpha'} \frac{1}{2} \chi_\pm^\mu = 0$$

$\leadsto$ These are second class, meaning that if we compute Poisson bracket the RHS is not another constraint.

$\hookrightarrow$ We replace the Poisson bracket by the Dirac Bracket

$$\{\chi_\pm^\mu(\sigma), \chi_\pm^\nu(\sigma')\} = -i\eta^{\mu\nu} 2\pi\alpha' \delta(\sigma - \sigma')$$

* Another method: Consider a Bosonic field $\varphi_i(t)$, $i=1, \dots, n$

whose action is

$$\tilde{\Lambda} = \int dt \left\{ \sum_{i=1}^n a_i(\varphi_i) \frac{d\varphi_i}{dt} - H(\varphi_i) \right\}$$

$$E.O.M: F_{ij} = \partial_i a_j - \partial_j a_i$$

$$F_{ij} \frac{d\varphi_i}{dt} = \partial H / \partial \varphi_i$$

$$0 \left(\frac{\partial \mathcal{L}}{\partial \varphi_i} \right) \frac{d\varphi_i}{dt} = \frac{\partial \mathcal{L}}{\partial \varphi_i}$$

$$\partial_a \left(\frac{\partial \mathcal{L}}{\partial (\partial_a \varphi^\mu)} \right) - \frac{\partial \mathcal{L}}{\partial \varphi^\mu} = 0$$

$$\begin{aligned} \partial \mathcal{L} / \partial \varphi^\mu &= \sum_i \partial_\mu a_i \frac{d\varphi_i}{dt} \cdot \delta_{\mu i} - \frac{\partial \mathcal{H}}{\partial \varphi^\mu} \delta_{\mu i} \\ &= \partial_j a_i \frac{d\varphi_i}{dt} \delta_{\mu j} - \partial \mathcal{H} / \partial \varphi^\mu \end{aligned}$$

$$\partial_\tau (a_\mu(\varphi_i)) = \frac{d}{dt} a_\mu = \frac{d\varphi_\ell}{d\tau} \frac{\partial}{\partial \varphi_\ell} a_\mu(\varphi)$$

$$\therefore \partial_\ell a_\mu \frac{d\varphi_\ell}{d\tau} - \partial_j a_i \frac{d\varphi_i}{d\tau} + \partial \mathcal{H} / \partial \varphi^\mu$$

$$\rightarrow F^{ij} F_{j\mu} = \delta_{\mu}^i$$

$$F^{ki} F_{ij} \frac{d\varphi_j}{dt} = F^{ki} \frac{\partial \mathcal{H}}{\partial \varphi_i} \Rightarrow \frac{d\varphi_i}{dt} = F^{ij} \partial_j \mathcal{H}$$

Which we compare to

$$d\varphi^i/dt = \{ \varphi^i, \mathcal{H} \} = \{ \varphi^i, \varphi^j \} \frac{\partial \mathcal{H}}{\partial \varphi^j}$$

$$\therefore \{ \varphi^i, \varphi^j \} = F^{ij}$$

* Now do the same to the free Majorana:

$$\mathcal{L} = \underline{P}^\mu \dot{\chi}_\mu - \mathcal{H}_\chi$$

$$(i/2) \bar{\chi}^\mu \gamma^0 \partial_0 \chi_\mu - i/2 \bar{\chi}^\mu \gamma^a \partial_a \chi^\nu \eta_{\mu\nu}$$

$$\frac{1}{4\pi\alpha'} \left((i/2) \bar{\chi}^\mu \gamma^0 \partial_0 \chi_\mu - (i/2) \bar{\chi}^\mu \gamma^a \partial_a \chi_\mu - i/2 \bar{\chi}^\mu \gamma^a \partial_a \chi_\mu \right)$$

$$\mathcal{H} = (i/2) \bar{\chi}^\mu \gamma^a \partial_a \chi_\mu (\eta_{\mu\nu})^{-1}$$

$$\begin{aligned} \therefore H &= \int d\sigma' \mathcal{H} = \frac{i}{4\pi\alpha'} \int d\sigma' \bar{\chi}^\mu(\sigma') \gamma^a \partial_a \chi_\nu(\sigma') \\ &= (-i/4\pi\alpha') \int d\sigma' \bar{\chi}^\mu(\sigma') \gamma^0 \gamma^1 \partial_1 \chi_\nu(\sigma') \end{aligned}$$

Therefore, the time derivative of χ_A^μ is given by

$$\dot{\chi}_A^\mu = \{ \chi_A^\mu, H \} = -\frac{2i}{4\pi\alpha'} \int d\sigma' \{ \chi_A^\mu(\sigma), \chi^\nu(\sigma') \} \gamma^0 \gamma^1 \partial_1 \chi_\nu(\sigma')$$

From the E.O.M:

$$\dot{\chi}^\mu = -\gamma^0 \gamma^1 \partial_1 \chi^\mu$$

$$\therefore \int d\sigma'' \left(\frac{i}{2\pi\alpha'} \{ \chi_A^\mu(\sigma), \chi^\nu(\sigma') \} \right) (-\gamma^0 \gamma^1 \partial_1 \chi_\nu(\sigma'))$$

$$\Rightarrow i/2\pi\alpha' \{ \chi_A^\mu(\sigma), \chi^\nu \} = \eta^{\mu\nu} \delta(\sigma - \sigma')$$

we conclude

$$\boxed{\{ \chi_A^\mu(\sigma), \chi^\nu(\sigma') \} = -2\pi\alpha' \eta^{\mu\nu} \delta(\sigma - \sigma')}$$

• We conclude $\boxed{\{ \psi^\mu(\sigma), \psi^\nu(\sigma') \} = -i\eta^{\mu\nu} 2\pi \delta(\sigma - \sigma')}$

$$\{ \alpha_n^\mu, \alpha_m^\nu \} = -i \eta^{\mu\nu} \delta_{n+m, 0}$$

$$\{ b_n^\mu, b_m^\nu \} = -i \eta^{\mu\nu} \delta_{n+m, 0}$$

$$\{ b_r^\mu, b_s^\nu \} = -i \eta^{\mu\nu} \delta_{r+s, 0}$$

* In the NS Sector we have:

$$G_r = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\sigma e^{ir\sigma} \sum_n \alpha_n^\mu (\tau) e^{-in\sigma} \sum_s b_{\mu s} e^{-is\sigma}$$

$$= \frac{1}{2\pi} \sum_n \sum_s \int_{-\pi}^{\pi} d\sigma e^{ir\sigma} e^{-in\sigma} e^{-is\sigma}$$

$$G_r = \frac{1}{2\pi} \sum_n (2\pi) \alpha_{r-s}^\mu \cdot b_{\mu s}$$

$$G_r = \sum_{n=-\infty}^{\infty} \alpha_{r+n}^\mu \cdot b_{\mu(-n)}$$

and

$$L_n = \frac{1}{2} \sum_{m=-\infty}^{\infty} \alpha_{n-m} \cdot \alpha_m + \frac{1}{2} \sum_{r=-\infty}^{\infty} b_{-r} \cdot b_{n+r} \left(r + \frac{n}{2} \right)$$

$\rho^\mu \rho_\mu$ was calculated for the Bosonic string,

$$\rightarrow i \Phi^\dagger \Delta \Psi^\mu \eta_{\mu\nu}$$

$$= \left(i \sum_s b_s^\mu e^{-is\sigma} \right) \left(\sum_t i s b_{t\mu} e^{-it\sigma} \right)$$

$$\therefore -\frac{1}{2\pi} \int_{-\pi}^{\pi} d\sigma e^{i\sigma} \left(- \sum_s \sum_t b_s^\mu s b_{t\mu} e^{is\sigma} e^{-it\sigma} \right)$$

$$= \sum_{s,t} \frac{1}{4\pi} \int_{-\pi}^{\pi} d\sigma e^{i\sigma(n-s-t)} s b_s \cdot b_t$$

$$= \frac{1}{2} \sum_s (b_s \cdot b_{n-s}) s$$

now $s \rightarrow -s$

$$= -\frac{1}{2} \sum_s b_{-s} \cdot b_{n+s} s$$

$$= -\frac{1}{2} \sum_s (\eta_{\mu\nu} b_{-s}^\mu b_{n+s}^\nu) s$$

$$= -\frac{1}{2} \sum_s \eta_{\mu\nu} \left[-b_{n+s}^\mu b_{-s}^\nu - i \eta^{\mu\nu} \delta_{n,0} \right] s$$

$$= \frac{1}{2} \sum_s b_{n+s} \cdot b_{-s} s + \frac{1}{2} n \delta_{n,0} \sum_s s$$

redundant

Note

$$\sum_{s=-\infty}^{\infty} s = 0 \text{ since } = \sum_{s=-\infty}^0 s + \sum_{s=2}^{\infty} s = \sum_{s=-\infty}^{-1} s + \sum_{s=1}^{\infty} s$$

Thus, alongside West's claim $\sum_{s=1}^{\infty} (-s) + \sum_{s=1}^{\infty} s = 0$

$\sum_r b_{n+r} \cdot b_{-r} = 0$ we can add this and see the above result.

* The algebra of Constraints becomes:

$$\begin{cases} \{L_n, L_m\} = -i(n-m)L_{n+m} \\ \{G_r, G_s\} = -2iL_{r+s} \end{cases} \quad \left| \quad \begin{cases} \{L_n, G_r\} = -i\left(\frac{n}{2} - r\right) G_{n+r} \end{cases}$$

$\Rightarrow \{L_n, L_m\}$ was done for the Bosonic String,

$$\begin{aligned} \{G_r, G_s\} &= \left\{ \sum_n \alpha_{-n} \cdot b_{r+n}, \sum_m \alpha_{-m} b_{s+m} \right\} \\ &= \alpha_n \cdot b_{r+n} \alpha_{-m} b_{s+m} + \alpha_{-m} b_{s+m} \alpha_{-n} \cdot b_{r+n} \\ &= \eta_{\mu\nu} \eta_{\lambda\sigma} \alpha_n^\mu (-\alpha_m^\sigma b_{r+n}^\nu) b_{s+m}^\lambda + 2-s \\ &= \sum_{n,m} (\eta_{\mu\nu} \eta_{\lambda\sigma}) (-\alpha_n^\mu \alpha_m^\sigma) (b_{r+n}^\nu b_{s+m}^\lambda) \\ &= \sum_{n,m} \eta_{\mu\nu} \eta_{\lambda\sigma} (\alpha_n^\mu \alpha_m^\sigma) [i\eta^{\mu\lambda} \delta_{r+n+s+m,0} + b_{s+m}^\lambda b_{r+n}^\mu] + \dots \end{aligned}$$

$$\begin{aligned} &= \sum_{n,m} \eta_{\mu\nu} \eta_{\lambda\sigma} \alpha_n^\mu \alpha_{-n}^\sigma \delta_{r+n+s+m,0} \quad -m = +n+r+s \\ &= \sum_n \alpha_{-n} \alpha_{n+r+s} + \sum \alpha \alpha b b = \underbrace{\alpha_{-n} \alpha_{-n} b_{s+n} b_{r+n}} \end{aligned}$$

$$\left(\sum_{n,m} (\alpha_{-n} \alpha_{-m} + i\eta^{\mu\nu} \delta_{n+m,0}) b_{s+n} b_{r+n} \right) \quad \leftarrow \text{sec. term}$$

$$\Rightarrow m = -n$$

$$\sum_n i(-n) b_{s-n} \cdot b_{r+n} + \alpha \alpha b b$$

$$= \sum_n (n) b_{s-n} b_{r+n} \quad r+n = s+m+s$$

Take $m \equiv n-s$

$$\therefore \sum_m (m+s) b_{-m} \cdot b_{r+s+m}$$

$$= \sum_m b_{-m} b_{r+s+m} (m+s)$$

A naive approach gets me close enough. I know there is some difficulties in the fact that $m, n, s, t \in \mathbb{Z} + 1/2$ so theres that. All will off by $\frac{s+t}{2}$ where $t/2$ is missing.

$$\{G_r, G_s\} = -2iL_{r+s} \quad \checkmark \dots \text{Kronecker } \delta$$

R-sector: $F_n = \sum_{m=-\infty}^{\infty} \alpha_{-m} \cdot d_{n+m}$

$$L_n = \frac{1}{2} \sum_{m=-\infty}^{\infty} \alpha \cdot m \cdot \alpha_{n+m} + \frac{1}{2} \sum_{m=-\infty}^{\infty} (2m+n) d_{-m} \cdot d_{m+n}$$

$$\{L_n, L_m\} = -i(n-m)L_{n+m}$$

$$\{L_n, F_m\} = -i(-m + \frac{n}{2}) F_{n+m}$$

$$\{F_n, F_m\} = -2i L_{n+m}$$

Closed Superstring:

$$\sigma \in [-\pi, \pi]$$

Thus, we get twice as many modes as for the open string together w/ a doubling of the constraints.

$$X^\mu(\tau, \sigma) = X^\mu(\tau, \sigma + 2\pi)$$

So, the B.C.'s involving this field automatically vanish:

$$\delta A / \delta (\partial_\sigma x^\mu) \quad \delta x^\mu \Big|_{-\pi}^{\pi} = 0$$

Boundary term: $\int_{\tau_1}^{\tau_2} d\tau \frac{\delta L}{\delta (\partial_\sigma x^\mu)} \delta x^\mu \Big|_{\sigma=-\pi}^{\sigma=\pi}$

$$\int_{\tau_1}^{\tau_2} d\tau \left(\partial_\sigma (X^\mu(\pi)) - \partial_\sigma X^\mu(-\pi) \right) \delta x^\mu \Big|_{-\pi}^{\pi}$$

Note $X^\mu(\tau, \sigma) = X^\mu(\tau, \sigma + 2\pi)$

so

$$\partial_\sigma X^\mu(\tau, \sigma) = \partial_\sigma X^\mu(\tau, \sigma + 2\pi)$$

(e.g. $\sigma = -\pi$, then

$$\partial_\sigma X^\mu(\tau, -\pi) = \partial_\sigma X^\mu(\tau, \pi)$$

→ So the closed string does not have B.C.'s:

$$\pi \tau_1 = \tau_2 \quad \pi \sigma_1 = \sigma_2 \quad \pi \tau_2 = \tau_1 \quad \pi \sigma_2 = \sigma_1$$

A the B.C. for the fermion is:

$$(\chi_{\pm}^{\mu} \delta \chi_{\pm\mu} - \chi_{\pm}^{\mu} \delta \chi_{\pm\mu})_{-\pi}^{\pi} = 0$$

Impose separately, B.C.'s on χ_{-}^{μ} and χ_{+}^{μ} and demand they are periodic or anti-periodic.

→ Each χ is independent

$$\begin{cases} \chi_{+}^{\mu}(\pi) = \chi_{+}^{\mu}(-\pi) & (R) \\ \chi_{+}^{\mu}(\pi) = -\chi_{+}^{\mu}(-\pi) & (NS) \end{cases}$$

$$\begin{cases} \chi_{-}^{\mu}(\pi) = \chi_{-}^{\mu}(-\pi) & (R) \\ \chi_{-}^{\mu}(\pi) = -\chi_{-}^{\mu}(-\pi) & (NS) \end{cases}$$

→ This gives four different possible combinations of B.C.'s:

$$(\chi_{+}, \chi_{-}) \in \{(R, R), (NS, R), (R, NS), (NS, NS)\}$$

(Possible set of B.C.'s)

* The constraints are exactly the same as the open-string:

$$T_{++} = 2\partial_{+}x^{\mu}\partial_{+}x_{\mu} - \frac{i}{2}\chi_{+}^{\mu}\partial_{+}\chi_{\mu+} = 0$$

$$T_{--} = 2\partial_{-}x^{\mu}\partial_{-}x_{\mu} - \frac{i}{2}\chi_{-}^{\mu}\partial_{-}\chi_{\mu-} = 0$$

$$J_{++} = 2\partial_{+}x^{\mu}\chi_{\mu+} = 0$$

$$J_{--} = 2\partial_{-}x^{\mu}\chi_{\mu-} = 0$$

(Closed String Constraints)

$$x^{\mu} = \sum_{n=-\infty}^{\infty} e^{in\sigma} \alpha_n^{\mu}$$

$$\dot{x}^{\mu} = \frac{1}{\sqrt{2\alpha'}} (\dot{x}^{\mu} + x^{\mu'}) = \frac{2}{\sqrt{2\alpha'}} \partial_{+} x^{\mu} = \sum_n \alpha_n^{\mu} e^{-in\sigma}$$

$$\ddot{x}^{\mu} = \frac{1}{\sqrt{2\alpha'}} (\ddot{x}^{\mu} - x^{\mu''}) = \frac{2}{\sqrt{2\alpha'}} \partial_{-} x^{\mu} = \sum_n \bar{\alpha}_n^{\mu} e^{in\sigma}$$

$$\{\alpha_n^{\mu}, \alpha_m^{\nu}\} = -in\delta_{n+m,0}\eta^{\mu\nu}$$

$$\{\bar{\alpha}_n^{\mu}, \bar{\alpha}_m^{\nu}\} = -in\delta_{n+m,0}\eta^{\mu\nu}$$

$$\chi_+^\mu = \sqrt{\alpha'} \sum_{n \in \mathbb{Z}} \alpha_n^\mu(\tau) e^{-in\sigma}$$

(R)

$$\chi_+^\mu = \sqrt{\alpha'} \sum_{r \in \mathbb{Z} + 1/2} b_r^\mu e^{-ir\sigma}$$

(NS)

$$\chi_-^\mu = \sqrt{\alpha'} \sum_{n \in \mathbb{Z}} \tilde{\alpha}_n^\mu(\tau) e^{+in\sigma}$$

(R)

$$\chi_-^\mu = \sqrt{\alpha'} \sum_{r \in \mathbb{Z} + 1/2} \tilde{b}_r^\mu(\tau) e^{ir\sigma}$$

(NS)

Since χ^μ satisfies

$$\partial \chi^\mu = 0$$

$$\chi_+^\mu = \sqrt{\alpha'} \sum_n \alpha_n^\mu |0\rangle e^{-in\sigma}$$

(R)

$$\chi_+^\mu = \sqrt{\alpha'} \sum_r b_r^\mu |0\rangle e^{ir\sigma}$$

(NS)

$$\chi_-^\mu = \sqrt{\alpha'} \sum_n \tilde{\alpha}_n^\mu |0\rangle e^{+in\sigma}$$

(R)

$$\chi_-^\mu = \sqrt{\alpha'} \sum_r \tilde{b}_r^\mu |0\rangle e^{ir\sigma}$$

(NS)

$$\{\chi_\pm^\mu(\sigma), \chi_\mp^\nu(\sigma')\} = -i\eta^{\mu\nu} 2\pi\alpha' \delta(\sigma - \sigma')$$

$$\{\alpha_n^\mu, \alpha_m^\nu\} = -i\eta^{\mu\nu} \delta_{n+m,0} \quad (\text{same for all others})$$

$$L_n = \frac{1}{2\pi\alpha'} \int_{-\pi}^{\pi} d\sigma e^{in\sigma} T_{++}$$

$$\bar{L}_n = \frac{1}{2\pi\alpha'} \int_{-\pi}^{\pi} d\sigma T_{--} e^{-in\sigma}$$

$$F_n = \frac{1}{\sqrt{2\alpha'}} \frac{1}{2\pi} \int_{-\pi}^{\pi} d\sigma e^{in\sigma} T_{++}$$

(R)

$$G_r = \frac{1}{\sqrt{2\alpha'}} \frac{1}{2\pi} \int_{-\pi}^{\pi} d\sigma e^{ir\sigma} T_{++}$$

(NS)

$$\bar{F}_n = \frac{1}{\sqrt{2\alpha'}} \frac{1}{2\pi} \int_{-\pi}^{\pi} d\sigma e^{-in\sigma} T_{--}$$

(R)

$$\bar{G}_r = \frac{1}{\sqrt{2\alpha'}} \frac{1}{2\pi} \int_{-\pi}^{\pi} d\sigma e^{ir\sigma} T_{--}$$

(NS)

In terms of oscillators:

NS-Sector:

$$L_n = \frac{1}{2} \sum_m \alpha_{n+m} \alpha_{-m} + \frac{1}{2} \sum_r (r + \frac{n}{2}) b_{-r} b_{n+r}$$

$$G_n = \sum_m \alpha_m \cdot b_{n-m}$$

R - Sector:

$$L_n = \frac{1}{2} \sum_m \alpha_{n+m} \alpha_{-m} + \frac{1}{2} \sum_m (m + \frac{n}{2}) d_{-m} \cdot d_{m+n}$$

$$F_n = \sum_m \alpha_m d_{n-m}$$

Claim: for the open-string: $\psi_+^M(l) = \pm \psi_-^M(l)$

$$\delta_\epsilon \psi_\pm^M = -\sqrt{\frac{2}{\alpha'}} \epsilon^\pm \partial_\pm \chi^\mu$$

Both NS and R B.C.'s breaks half the worldsheet

$$\text{susy: } \psi^+(0) = \psi^-(0)$$

$$\psi^+(\pi) = \pm \psi^-(\pi)$$

Recap: Brink-Schwarz Superparticle:

• Spacetime SUSY

$$X^\mu, \mu = 0, \dots, D-1,$$

$$\Theta_A: \text{Spacetime Majorana} \mid \bar{\Theta}^M = \Theta^T C = \bar{\Theta}^D$$

$$A = 1, \dots, 2^{\lfloor D/2 \rfloor}$$

Susy:

$$\delta X^\mu = i \bar{\epsilon} \gamma^\mu \Theta, \quad \delta \Theta_A = \epsilon_A$$

ϵ_A : constant space-time Majorana

$$[\delta_1, \delta_2] X^\mu \sim i \bar{\epsilon}_2 \gamma^\mu \epsilon_1$$

commutator of two susys is a spacetime translation.

$$\bar{\lambda} \gamma^\mu \rho = - \bar{\rho} \gamma^\mu \lambda$$

$$T^\mu \equiv \dot{X}^\mu - i \bar{\Theta} \gamma^\mu \dot{\Theta}$$

is invariant under susy:

$$\begin{aligned} \delta T^\mu &= \frac{d}{dt} (i \bar{\epsilon} \gamma^\mu \Theta) - i \bar{\epsilon} \gamma^\mu \dot{\Theta} - i \dot{\bar{\Theta}} \gamma^\mu \Theta \\ &= i \bar{\epsilon} \gamma^\mu \ddot{\Theta} - i \ddot{\bar{\Theta}} \gamma^\mu \Theta \end{aligned}$$

T^μ transforms in the usual way under Lorentz

Symplectic transformations

$$\delta x^m = \omega^m_n x^n, \quad \delta \theta = \frac{1}{4} \omega_{mn} \gamma^{mn} \theta$$

so π^m transforming as a vector:

$$\delta \theta^T = \frac{1}{4} \omega_{mn} \theta^T (\gamma^{mn})^T$$

$$\begin{aligned} \delta \bar{\theta} &= \frac{1}{4} \omega_{mn} (\theta^T \gamma^0) (\gamma^0 (\gamma^{mn})^T \gamma^0) \\ &= \frac{1}{4} \omega_{mn} \bar{\theta} (\gamma^0 (\gamma^{mn})^T \gamma^0) \end{aligned}$$

* For a massless pt. particle: to have it be spacetime supersymmetric: Replace x^m by π^m :

$$A = \frac{1}{2} \int d\tau e^{-1} \pi^m \pi^n \eta_{mn}$$

Since $e = e(\tau)$ it is next to susy (its also an E.O.M. Σ)

$$\begin{aligned} \frac{d}{d\tau}(e^{-1} \pi^m) &= 0, & \pi \dot{\theta} &= 0, & \pi^m \pi^n \eta_{mn} &= 0 \\ (x^m) & & (\theta) & & (e) & \end{aligned}$$

Conjugate momenta: $P^m = e^{-1} \pi^m \Leftrightarrow P_m = e^{-1} \pi_m$
the Action becomes:

$$A = \int d\tau (\pi^m \dot{x}^m - \frac{1}{2} e p^m p_m) \eta_{mn}$$

Here A is invariant under susy: $\delta e = 0, \delta p^m = 0$

... can see more about this theory. It possess an extra symmetry called κ -symmetry.

* Brink-Schwarz Super String:

$$x^m(\tau, \sigma), \theta^A i, \quad A = 1, \dots, 2^{\lfloor D/2 \rfloor}, \quad i = 1, 2$$

i : means this is $N=2$ susy.

$$\delta \theta^i = \epsilon^i$$

$$\delta x^m = i \bar{\epsilon}^j \gamma^m \theta_j \quad | \quad \epsilon \text{ are constant}$$

We define: $\pi^m_\alpha = \partial_\alpha x^m - i \bar{\theta}^j \gamma^m \partial_\alpha \theta_j$

which is invariant under susy

proof:
$$\begin{aligned} \delta \pi^m_\alpha &= \partial_\alpha (\delta x^m) - i \delta \bar{\theta}^j \gamma^m \partial_\alpha \theta_j - \bar{\theta}^j \gamma^m \partial_\alpha (\delta \theta_j) \\ &= \partial_\alpha (i \bar{\epsilon}^j \gamma^m \theta_j) - i \bar{\epsilon}^j \gamma^m \partial_\alpha \theta_j - \bar{\theta}^j \gamma^m \partial_\alpha (\epsilon_j) = 0 \end{aligned}$$

$$L = i\bar{E}\gamma^{\mu}\partial_{\alpha}\Theta^i - i\bar{\Theta}\gamma^{\mu}\partial_{\alpha}\Theta^i = 0$$

Thus, as with the pt-particle $\partial_{\alpha}x^{\mu} \rightarrow \pi^{\mu}_{\alpha}$
 so Polyakov action becomes:

$$S_1 = -\frac{1}{2\pi} \int d^2\xi \sqrt{-g} g^{\alpha\beta} \pi^{\mu}_{\alpha} \pi^{\nu}_{\beta} \eta_{\mu\nu}$$

- This theory does not have K-symmetry which the pt-particle has. This symm is required in order to get the correct on-shell D.O.F. for Θ^i .

To recover this symm we need to add

$$S_2 = \frac{1}{\pi} \int d^2\xi \epsilon^{\alpha\beta} \{ i \partial_{\alpha} x^{\mu} (\bar{\Theta}^1 \gamma^{\mu} \partial_{\beta} \Theta^1 - \bar{\Theta}^2 \gamma^{\mu} \partial_{\beta} \Theta^2) \\ + \bar{\Theta}^1 \gamma^{\mu} \partial_{\alpha} \Theta^1 \bar{\Theta}^2 \gamma^{\mu} \partial_{\beta} \Theta^2 \} \eta_{\mu\nu}$$

$S = S_1 + S_2$ is Lorentz and reparametrization